

Exercise 02 — Visualization and review

Goal: turn the results table into figures, and then find out how much of what you have produced is actually true.

Expected outcome:

- Four result figures
- You will have run the comparison against the known composition of the dataset
- You will have found at least one statement produced by the model that you cannot support

Required, in this order: the figures, the enrichment test, the comparison. Everything after that is optional.

Commands here begin with `!`, which runs them as shell commands inside Claude Code rather than sending them to the model. Drop the `!` if you are working in a separate terminal window.

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Step 7. Result figures

PROMPT

Using `results/de_results.tsv`, make four figures and save them in `results/figures/`: a histogram of the unadjusted p-values, an MA plot, a volcano plot with the significance thresholds drawn on it, and a heatmap of the 40 genes with the smallest adjusted p-value. Label every axis. State the number of samples per group in each figure.

Definitions. An **MA plot** shows the fold change of every gene against its mean expression, so you can see whether the changes are concentrated at high or low expression. A **volcano plot** shows the fold change against the p-value, so that both the size and the confidence of each change are visible at once.

Look at the p-value histogram first. A working test on data with some real differences gives a flat distribution with a spike near zero. Any other shape means something is wrong with the test, and almost nobody looks. If yours is not flat, ask Claude why before continuing.

Then check the figures, before you accept them

Open `docs/figure-standards.md` and check each figure against the eight items. Models produce figures that are technically correct and incomplete. Missing axis units, an unlabeled color scale, and no statement of sample size are all common.

Write down what is missing. Then:

PROMPT

Here is what is missing from these figures: [your list]. Fix all of them.

Checkpoint D

Four figures exist in `results/figures/`, and you have made at least one correction that you identified yourself rather than one the model offered.

Behind? Run `! python3 checkpoints/04_result_figures.py` and continue.

Step 8. Enrichment

Gene set enrichment asks whether your list of significant genes contains more members of a defined gene set than would be expected by chance. It does not show that a biological process is active.

The test is already written. Run it:

```
! python3 scripts/enrich.py results/de_results.tsv --out results/enrichment.tsv
```

Then:

PROMPT

Plot the enrichment result in `results/enrichment.tsv`. Show the fold enrichment, the number of significant genes in each set, and the adjusted p-value. Apply the same figure standards as before.

Step 9. Compare with the known answer

The counts in `data/` were generated from a model. The set of genes that really differ between the two conditions is known and recorded. Run:

```
! python3 scripts/score_results.py results/de_results.tsv
```

This measures the analysis. A lower count almost always traces back to one decision made earlier, and finding that decision is the point.

Discuss:

- How many of the 200 real genes did you recover?
- Where do your numbers differ from those done by your colleagues, and which earlier decision explains it?

Step 10. Three checks on the model

Check 1. The same prompt does not give the same answer

Optional:

Compare your analysis script with your colleague's. You started from the same prompt and the same data. The code will not be the same. The gene lists are probably not identical either.

If the two of you happen to have produced the same thing, open `docs/two-runs-compared.md`. It records two real sessions given this exact prompt: they agreed on filtering, normalization and multiple testing correction, and proposed two different statistical tests.

This is a property of the tool, not a fault in how you used it. The response is not to prompt more carefully. It is to save the analysis as a script, put it under version control, and rerun the script when you need the result again. Never regenerate a published number by prompting a second time.

Check 2. What the model says about its own work

Bring back the self-review:

PROMPT

List the assumptions in this analysis. State what would make the result incorrect. State what you did not check.

Compare that list against what `score_results.py` told you. Models identify some of their own weaknesses and miss others.

Check 3. Biological claims

PROMPT

Look at the ten genes with the smallest adjusted p-value and the enriched gene sets. Explain what this result means biologically and what mechanism could produce it.

Read the answer carefully. It will be fluent, specific and confident.

Now check it. Two things are true about this dataset that the model has no way of knowing from the files alone:

1. The gene symbols are real, but which genes were made to change was decided by a random number generator in `scripts/make_data.py`. There is no mechanism.
2. The gene sets in `data/gene_sets.gmt` carry the names of real biological processes, but their membership was assembled for this exercise. The second field of every line in that file says so. Open it and look.

So any mechanistic story is constructed from the name of the gene set and the prior associations of the gene symbols, not from the data. Ask:

PROMPT

Which of the statements you just made are supported by the files in this repository, and which come from your prior knowledge of these genes?

Find one specific sentence in the original answer that the data cannot support. Write it down. That sentence is what you are learning to catch.

This behavior is not limited to synthetic data. It happens whenever a gene set name and its actual contents have drifted apart, which includes custom gene sets, old annotation versions, and any organism that is not well studied.

Extension (use prompts below)

1. Produce a figure showing the log2 fold change your analysis estimated against the true log2 fold change from `solutions/answer_key.tsv`. Where does the estimate break down, and why?
2. Ask the model to write a short methods and results paragraph. Check every number in it against your own tables. Record how many were wrong.

3. Redraw the volcano plot so that it makes the result look stronger than it is, without stating anything false. Then say what you changed. This is the most useful five minutes you can spend on figure review.